E0X Adiabatic Batch Reactor Example

The irreversible, elementary liquid phase reaction A + B --> C is carried out adiabatically in a 10-L batch reactor. Species A and B are present initially in equimolar amounts. Determine the time needed to achieve 90% conversion and the resulting temperature. T0 = 300 K.

CA0 = 0.1 mol/L
CPA = CPB = 0.015 kcal/mol/K
CPC = 0.030 kcal/mol/K
$\Delta H_{f,A}^{0} = -20\ \mathrm{kcal/mol}$
$\Delta H_{f,B}^{0} = -15\ \mathrm{kcal/mol}$
$\Delta H_{f,C}^{0} = -41\ \mathrm{kcal/mol}$
Eact = 10 kcal/mol
k = 0.01 L/mol/s at 325 K

Mole balance: $N_{A0}\frac{dX}{dt} = -r_A V$

liquid phase so V is constant, so

can simplify to $C_{A0}\frac{dX}{dt} = -r_A$, $C_{A0} = 0.1\ \frac{mol}{L}$

rate law: $-r_A = k C_A C_B$

Concentration eqs:

$$C_A = C_B = C_{A0}(1-X)$$

$$\therefore\ -r_A = k C_{A0}^2 (1-X)^2$$

Arrhenius eq:

$$k = \left(0.01\ \frac{L}{mol \cdot s}\right) \exp\left[\left(\frac{10\ \frac{kcal}{mol}}{0.001986\ \frac{kcal}{mol \cdot K}}\right)\left(\frac{1}{325\ K} - \frac{1}{T}\right)\right]$$

Energy balance:

2 ways, either is fine:

(1) $$T = \frac{T_0 \sum \Theta_i C_{pi} + X(-\Delta H_{rxn} + \Delta C_p \cdot 298.15K)}{\sum \Theta_i C_{pi} + X \Delta C_p}$$

terms:

→ $\Delta H_{rxn} = \Delta H^\circ_{f,C} - \Delta H^\circ_{f,A} - \Delta H^\circ_{f,B} = -b \frac{kcal}{mol}$

→ $\sum \Theta_i C_{pi} = \Theta_A C_{pA} + \Theta_B C_{pB} + \Theta_C C_{pB}$

$= (1)(0.015 \frac{kcal}{mol \cdot K}) + (1)(0.015 \frac{kcal}{mol \cdot K}) + (0)(0.03 \frac{kcal}{mol \cdot K})$

$= 0.030 \frac{kcal}{mol \cdot K}$

→ $\Delta C_p = \sum_{products} \left|\frac{\nu_i}{\nu_A}\right| C_{pi} - \sum_{reactants} \left|\frac{\nu_i}{\nu_A}\right| C_{pi}$

$= C_{pC} - C_{pA} - C_{pB} =$

$= 0.030 \frac{kcal}{mol \cdot K} - 0.015 \frac{kcal}{mol \cdot K} - 0.015 \frac{kcal}{mol \cdot K}$

$\Delta C_p = 0$

$$\therefore T = \frac{(300K)(0.030 \frac{kcal}{mol \cdot K}) + X (6 \frac{kcal}{mol})}{0.030 \frac{kcal}{mol \cdot K}}$$

- solve simultaneously

<u>way ②</u>

use the general EB:

$$\frac{dT}{dt} = \frac{(\Delta H_{rxn})(r_A V)}{N_{A0}(\Sigma \Theta_i C_{pi})} = \frac{(-6 \frac{kcal}{mol})(r_A)}{C_{A0}(0.030 \frac{kcal}{mol \cdot K})}$$

- where $C_{A0} = 0.1 \frac{mol}{L}$ and

$r_A = -k C_{A0}^2 (1-X)^2$ and

k is the function of T from above

couple the EB w/ the MB
& solve simultaneously

NOTE: Since X & T both vary with time (t), you must solve simultaneously.

PFR & batch reactor require simultaneous solution even for an adiabatic reactor, CSTR if you have an adiabatic reactor, you can solve sequentially (e.g. given X, solve for T, use T to get k (and K_c, C_i, etc as needed), use those to solve your mole balance.)